



Weld seam detection based on 2D image and 3D point cloud segmentation: a coarse-to-fine weld path planning method

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Received: 27 February 2025 / Accepted: 5 November 2025 / Published online: 22 November 2025
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Abstract

The path planning of welding robots is a critical step toward achieving high-quality robotic welding. Automatic seam detection and path planning have emerged as a prominent research hotspot. This study proposed a method for automatic robotic welding path planning. First, an efficient deep neural network (denoted as EfficientWeld) is developed for weld seam detection and endpoint localization, with the objectives of achieving gross localization of weld seam regions and guiding subsequent precise scanning. The EfficientWeld model demonstrates accurate extraction of start and end points for butt joints, lap joints, and irregular joints, achieves a 36% improvement in inference speed, and exhibits robust performance in scenarios involving multiple weld seams and varying backgrounds. Subsequently, the identified joint start and end points are utilized to guide structured laser sensors in performing fine scanning of workpieces and extraction of weld seams. To accelerate feature extraction, an enhanced gray gravity method is proposed. Finally, a spatial position and pose model for complex curved weld seams is constructed using 3D point cloud data, enabling high-precision path planning and robotic welding posture planning for complex curved seams. The root mean square error (RMSE) of the generated path is less than 0.25 mm, with a maximum lateral error below 0.61 mm, satisfying the requirements for practical industrial applications.

Keywords Weld seam detection · Point cloud segmentation · Weld path planning · Deep learning

1 Introduction

Welding technology constitutes a critical component of intelligent manufacturing, driving increasing demand for automated and intelligent welding robots. Currently, the “teach-playback” mode and offline programming (OLP) remain the two dominant operational methods in modern manufacturing [1–3]. However, the “teach-playback”

mode relies on manual point-by-point guidance to record robot trajectories, yet it suffers from inherent drawbacks such as low efficiency and lengthy production preparation time. When addressing complex weld seams or workpiece deviations, it demands repeated adjustments to robot orientation, speed, and welding parameters. Critically, the fixed paths generated by this method are only compatible with the workpieces and environments present during the programming phase. Consequently, any modifications to workpiece design, clamping setups, or environmental conditions necessitate complete reprogramming. As modern manufacturing models evolve, unstructured welding environments [4] are emerging continuously; thus, the development of intelligent welding robots capable of adapting to novel, unstructured, and flexible manufacturing paradigms is imperative.

Offline programming involves analyzing the workpiece’s theoretical model to mathematically derive the geometric information of the groove to be filled. On this basis, the robot’s welding trajectory is planned, and the welding process is simulated [4, 5]. Offline programming substantially

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enhances welding efficiency by reducing the manual teaching workload. Liu et al. [6] achieved efficient planning of all welding path points for large grooves by the initial weld pass offline programming and an offset strategy. Liu et al. [7] proposed a robotic trajectory offline programming method for welding multi-tube intersecting structures in the form of a common main pipe, which enhances welding quality and efficiency. However, the accuracy of offline programming is inherently constrained by the precision of the theoretical model [8]. In production environments, deviations between actual dimensions and the theoretical model frequently occur owing to limitations in weld processing accuracy and welding-induced deformation. In such scenarios, adjustment or regeneration of the welding path is required [9–11]. Furthermore, many welding structures lack corresponding theoretical models, necessitating weld seam identification and automatic path planning.

Currently, various sensors, including arc sensors [12], photodiodes [13], infrared sensors [14], and vision sensors [15–18], have been employed for weld seam identification and path planning in intelligent welding robots. Compared to other sensor types, vision sensors capture rich environmental information and offer high precision. Vision sensors are categorized into passive optical and active optical. Active optical vision sensors exhibit higher accuracy and robust performance, making them more widely adopted in industrial measurement fields such as assembly, welding, and inspection [19–23].

Structured light vision serves as a key representative of active optical vision. Laser structured light is primarily utilized for real-time weld seam tracking and identification. However, it functions as a localized sensor incapable of perceiving global environments [24]. As a result, operators typically must provide approximate guidance for the scanning path. The continuously varying curvature of curved weld seams renders welding posture teaching both difficult and time-consuming. Furthermore, weld seam tracking is generally applicable only when workpiece positional changes are minor. For significant workpiece positional changes, manual alignment of the welding torch with the initial point is required prior to welding, thereby reducing robotic welding efficiency.

To minimize human intervention in the welding process and enhance welding automation levels, various 3D vision sensors have been deployed for welding path generation. Huang et al. [25] and Wang et al. [26] employed stereo vision sensors to capture workpiece images and achieved automatic welding path extraction via baseline correction, stereo matching, and 3D reconstruction. However, in welding environments with complex backgrounds and sparse textures, stereo vision-based methods encounter challenges including poor feature point matching and low detection accuracy.

Kim et al. [27] and Zhou et al. [28] utilized RGB-D sensors to acquire workpiece point clouds and performed weld seam detection and path generation using edge detection methods. Nevertheless, the positioning accuracy of RGB-D sensors warrants further improvement, as it directly impacts the precision of welding path generation. Zhang et al. [29] and Liu et al. [30] developed laser structured light sensors to capture workpiece point clouds, from which weld seam feature points were extracted for automatic welding path generation. However, the efficiency of automatic welding path generation is compromised by the manual teaching required for scanning weld workpieces.

With the advancement of artificial intelligence, deep learning has been integrated into the weld identification process. Ma et al. [31] applied AI algorithms to welding feature point recognition and achieved favorable results. Liu et al. [32] leveraged deep learning's feature extraction capabilities to predict key nodes and weld seam positions. Chen et al. [33] proposed methods for automatic weld seam type classification, stud point recognition, and welding seam region-of-interest (ROI) determination based on YOLOv5. These studies demonstrate that applying deep learning to welding workpiece identification and using it to detect the approximate position and orientation of weld seams—rather than relying on manual scanning path teaching—can further enhance the intelligence of the welding process.

To address the limitations of existing methods in weld seam extraction, a deep learning-based path planning method is proposed. The contributions of this work are summarized as follows:

- 1) The method adheres to a coarse-to-fine paradigm, ensuring both accuracy and efficiency in weld seam extraction. Artificial intelligence is employed to identify the approximate weld position, guiding the scanner to perform local fine scanning—an approach that circumvents the limitations of global scanning, including excessive data volume and feature extraction difficulties.
- 2) A deep learning model named EfficientWeld is proposed to rapidly and robustly extract weld seams and their endpoints for diverse weld types. This model exhibits strong feature extraction capabilities and is applicable to lap welds, butt welds, and irregular welds.
- 3) A point cloud data-based welding path planning method is proposed, enabling high-precision path planning and robotic welding posture planning for complex curved seams.

The remainder of this article is structured as follows: Section II describes the experimental system and vision model; Section III presents the weld path planning framework; Section IV details the experiments and results; and Section V concludes the article.

2 Experimental system and vision model

2.1 Experimental system and vision sensor

Figure 1 shows the experimental platform setup, which comprises a TIG torch, a wire feeder, and a vision sensor. The vision sensor integrates an industrial camera and a structured-light laser emitting at 450 nm. The laser is mounted to project vertically onto the welding platform, and a fixed angular offset between the industrial camera and the laser facilitates the application of the laser triangulation measurement method. The TIG torch and vision sensor are both mounted on a FANUC 6-axis robot. The camera, robot manipulator, and vision computer are interconnected via Ethernet cables to a common switch, enabling high-speed data transmission and command execution within a local area network (LAN). The robotic manipulator employed in this work is a FANUC M-20iB, whose key technical specifications are listed in Table 1. The vision computer is configured with an Intel Core i7-10750 H processor, an NVIDIA GTX 1650 Ti GPU, 16 GB of random-access memory (RAM), and a 1 TB solid-state drive (SSD). Additionally, a computer equipped with an Intel Core i7-7900X processor, 32GB of RAM, and an NVIDIA RTX 3090 graphics card is dedicated to training deep neural network models.

TIG welding is commonly employed in applications where stricter welding quality is required. Owing to the often requirement for supplementary wire feeding in TIG welding, path planning for TIG processes requires the explicit consideration of the wire feeding angle—a constraint that imposes greater limitations on torch rotation. Consequently, the path planning complexity of TIG welding is higher than that of MAG/MIG welding. Notably, no fundamental differences exist in the workpiece geometries applicable to both TIG welding and MAG/MIG welding. Therefore, the proposed method can be equally applied to MAG/MIG welding processes.

Table 1 Robot technical specifications

Case	Parameters	Value
1	Number of robot axes	6
2	Repeat positioning accuracy (mm)	0.02
3	Maximum load at end (kg)	25
4	Maximum reach radius (mm)	2032

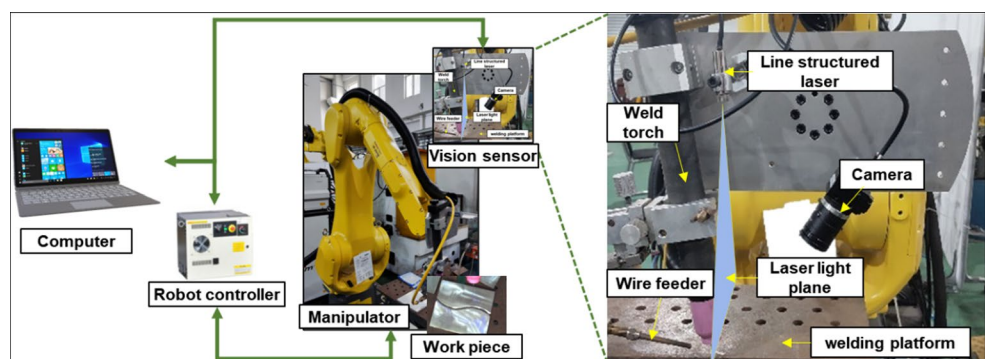
2.2 Coordinate system definition

The world coordinate system (O_W) serves as the reference frame for the positioning system. Any spatial point can be localized and described within this framework. The camera coordinate system (O_c) is typically defined with its origin at the optical center of the camera. The X_C and Y_C axes of the camera coordinate system are parallel to the x- and y-axes of the image plane. The direction of the Z_C axis, determined by the right-hand rule, is consistently perpendicular to the image plane. The robot end-effector coordinate system (O_E) corresponds to the coordinate frame of the robot's end-effector. Motion planning in the robot control system is typically based on the position and orientation of this frame. The tool coordinate system (O_t) is defined with the tool center point (TCP) as its origin and varies with tool motion.

Once the hardware system is constructed, the relative transformations between the coordinate systems are established. As illustrated in Fig. 2, O_c , O_t , O_E , and O_W correspond to the camera, tool, robot end-effector, and world coordinate systems, respectively. Transformation matrices t_cT , E_tT , and W_ET denote the transformations from O_c to O_t , O_t to O_E , and O_E to O_W , respectively. A point in the camera coordinate system (P_{O_c}) can be transformed to the world coordinate system (P_{O_w}) using the following equation:

$$P_{O_w} = {}^W_E T {}^E_t T {}^t_c T P_{O_c} \quad (1)$$

Fig. 1 System platform and vision sensor



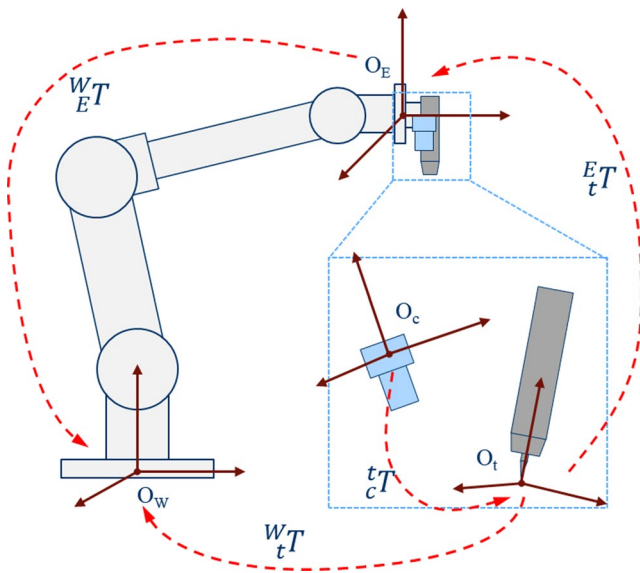


Fig. 2 The relative positional relationships between different coordinate systems

Here, $W_E^T E_t^T$ can be simplified to W_t^T , which is obtained via the teach pendant following TCP calibration. t_c^T is obtained through hand-eye calibration.

2.3 Line structured laser vision model

The pinhole model describes the mapping of 3D world coordinate points to the 2D image plane by a camera. According to this model, the projection of a world coordinate point $P_w = [x_w, y_w, z_w, 1]^T$ onto the image plane $P_{img} = [u, v, 1]^T$ is expressed as:

$$z_C \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = AEM \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix} \quad (2)$$

where the matrices are defined as:

$$A = \begin{bmatrix} 1/dx & 0 & u_0 \\ 0 & 1/dy & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad (3)$$

$$M = \begin{bmatrix} R(\varphi, \theta, \phi) & t \\ 0 & 1 \end{bmatrix} \quad (4)$$

Where z_C is scale factor, A is the camera intrinsic parameter matrix, M is the camera exterior parameter matrix, $R(\varphi, \theta, \phi)$, t is the rotation matrix and translation vector from the global coordinate to the camera coordinate. E is the 4×4 identity matrix. The camera intrinsic parameters were calibrated using Zhang's method [34].

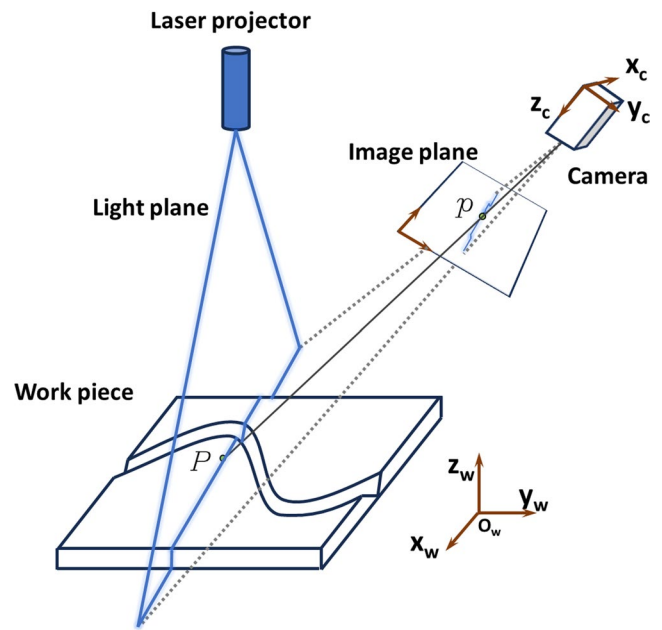


Fig. 3 Geometric structure of line structured laser vision sensor model

The world coordinate system was established with its XOY plane aligned to the upper surface of the welding platform and the Z-axis perpendicular to this plane. The transformation matrix $t^W T$ was then automatically calculated by the robot.

As shown in Fig. 2, for any two robot poses during motion, the following Eq. (5) holds:

$$W_E^T E_t^T t_c^T {}_b^c T_1 = W_E^T E_t^T t_c^T {}_b^c T_2 \quad (5)$$

where $W_E^T E_t^T = W_t^T$, which can be obtained from the robot teach pendant. ${}_b^c T_i$ denotes the extrinsic matrix of the calibration target, calculated via Zhang's Camera Calibration Algorithm [34].

After multiple calibrations, the transformation matrix t_c^T with the minimal error was selected using Eq. (5). The point cloud data P_W in the robot coordinate system was then derived from the vision sensor's pose and intrinsic matrix.

Consider a world coordinate point $P = [x_w, y_w, z_w, 1]^T$ projecting to image point $p = [u, v, 1]^T$ (Fig. 3). This point P also lies on the light plane associated with the camera, whose global coordinate equation is:

$$ax_w + by_w + cz_w + d = 0 \quad (6)$$

where a, b, c and d represent the four coefficients of the light plane equation.

In the global coordinate system, the ray Op is defined by the camera model, and the light plane by Eq. (6). Thus, the 3D global coordinates of P are uniquely determined by the intersection of Op and the light plane. Point P_{xyz} satisfies

both Eqs. (1) and (6). Combining these equations yields the x, y, z coordinates corresponding to each (u, v) pair.

3 Framework for weld path planning

The proposed method adopts a coarse-to-fine paradigm to localize welds on the workpiece. As illustrated in Fig. 4, the overall workflow comprises the following four steps:

- 1) **Camera Calibration and Initial Detection:** Camera intrinsic calibration and hand-eye calibration are first performed. Workpiece images are then captured, and the EfficientWeld network is employed to detect weld seam regions and endpoint positions.
- 2) **Coordinate Transformation:** The boundary coordinates of the weld seam region and the weld seam endpoint coordinates in the pixel coordinate system are obtained. These coordinates are transformed into the robot coordinate system using camera calibration data.
- 3) **Scanning Path Generation and Point Cloud Acquisition:** A scanning path is generated based on the weld seam endpoint coordinates. Along this path, detailed

scanning of the weld seam's 3D morphology is conducted, yielding 3D point clouds of both the weld seam and the workpiece.

- 4) **Point Cloud Segmentation and Welding Execution:** The point cloud is segmented using the boundary coordinates of the weld seam region. Weld seam coordinates are extracted from the segmented 3D point cloud, and welding torch postures are generated. Finally, the robot is controlled for welding based on the weld seam coordinates and torch postures.

3.1 Welding seam detection network

An efficient deep neural network, termed EfficientWeld, is proposed for accurate identification of weld locations and endpoints within images. The tasks of weld seam recognition and endpoint localization are formulated as an object detection problem. A one-stage detection framework is introduced, which simplifies the detection process into a single neural network inference step to directly predict target bounding boxes and classification results. Finally, EfficientWeld outputs the pixel coordinates of weld endpoints and the weld border within the image.

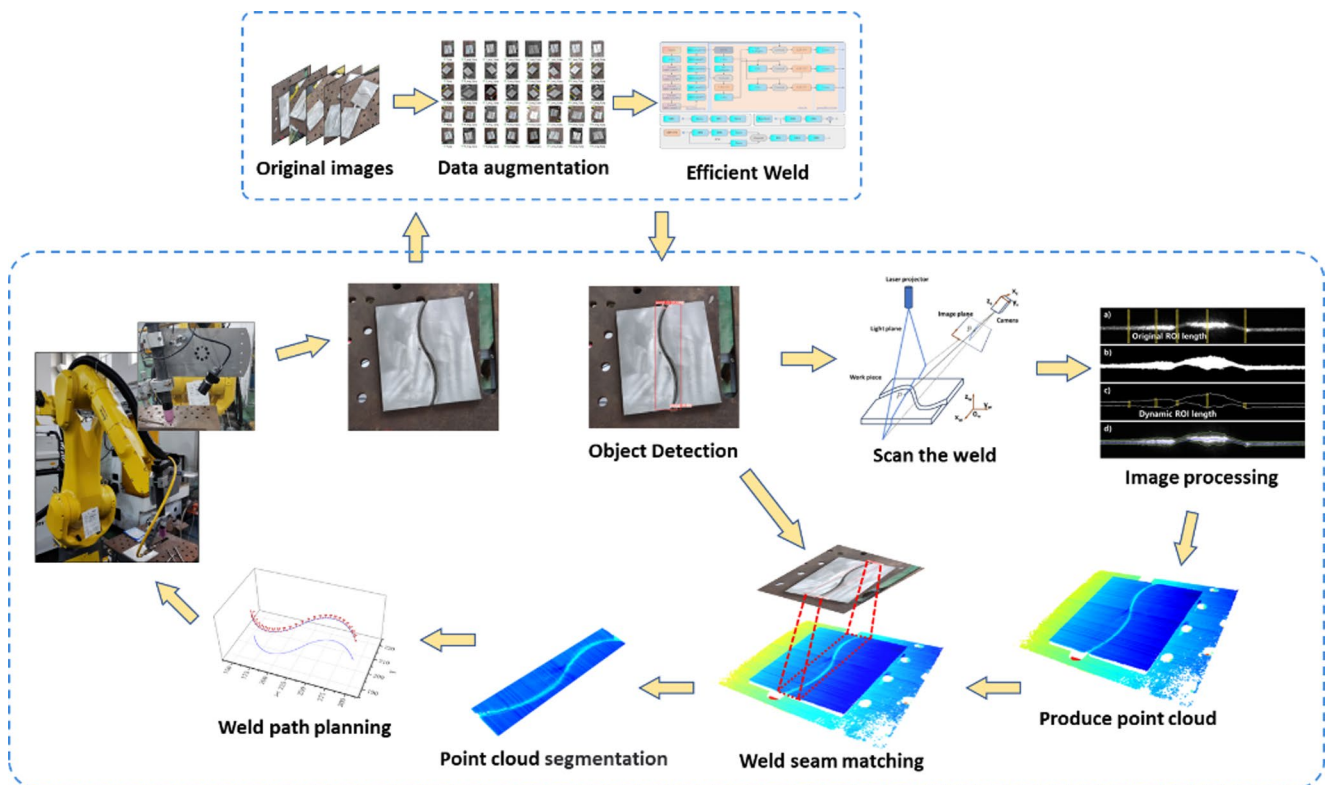


Fig. 4 Framework of welding path automatic generation

3.1.1 Network structure

As illustrated in Fig. 5, the EfficientWeld model comprises four principal components: input, backbone, neck, and prediction modules. The backbone module extracts image features, including MBConv and FusedMBConv modules. The MBConv module serves as the core component of the MobileNet [35] series. It employs depthwise separable convolution to reduce parameter count and computational cost, achieving high feature extraction capability with minimal computational overhead—making it suitable for lightweight models. FusedMBConv, a module introduced in EfficientNetV2 [36], replaces depthwise separable convolution with standard convolution. Standard convolution exhibits stronger global feature capture capability when processing small-sized input features, thereby improving the quality of shallow features. A hybrid approach combining MBConv and FusedMBConv modules is adopted. FusedMBConv is used in shallow layers to accelerate training and optimize feature extraction, whereas MBConv is employed in deeper layers for efficient processing of more complex features—this reduces computational cost. The neck module integrates features extracted from the backbone module to further enhance network performance. The prediction module leverages fused features for target prediction. Notably, the CBS module comprises convolution (Conv), batch normalization (BN), and SiLU activation functions. The CSP2*X module incorporates CBS, Conv, and bottleneck

layers, further enhancing the network's feature integration capability.

3.1.2 Loss function

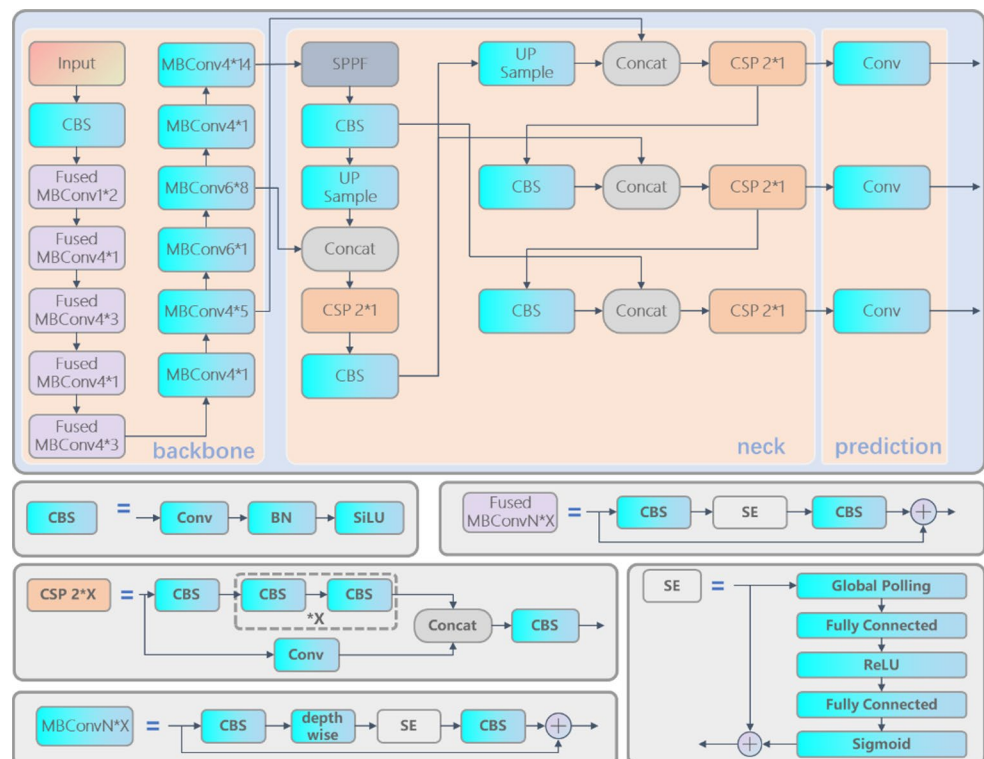
A well-designed loss function effectively measures sample similarity and accelerates model convergence. The total loss function of EfficientWeld is defined as:

$$\mathcal{L} = e_{box} * \mathcal{L}_{box} + e_{conf} * \mathcal{L}_{conf} + e_{cls} * \mathcal{L}_{cls}$$

where $\mathcal{L}$ denotes the total loss; $\mathcal{L}_{box}$, $\mathcal{L}_{conf}$ and $\mathcal{L}_{cls}$ represent the localization loss, confidence loss, and classification loss, respectively; and e_{box} , e_{conf} and e_{cls} are the weighting coefficients for each loss component relative to the total loss.

In object detection tasks, Intersection over Union (IoU) is commonly employed as a loss function to evaluate model performance. However, IoU suffers from inherent limitations: it only accounts for overlap between two bounding boxes, neglecting both the distance between their center points and shape matching. Two target classes are investigated in this study: accurate localization of weld seam end-points is more critical, whereas the shape integrity of weld seam regions is more important. Thus, the Complete IoU (CIoU) [37] function is adopted for localization loss calculation. CIoU incorporates a penalty term for center point distance, encouraging the predicted box center to align more

Fig. 5 The structure of the EfficientWeld model



closely with the target box center. It also addresses shape matching by incorporating aspect ratio similarity.

$$\mathcal{L}_{box} = 1 - IOU + \frac{d^2}{c^2} + \alpha v \quad (7)$$

where d is the distance between the centers of the predicted and true boxes, and c is the diagonal distance of the minimum enclosing rectangle.

$$\alpha = \frac{v}{(1-IOU)+v} \quad (8)$$

$$v = \frac{4 \left(\arctan \frac{w_G}{h_G} - \arctan \frac{w_P}{h_P} \right)^2}{\pi^2} \quad (9)$$

where w_G , h_G , w_P , h_P denote the width and height of the true box and the predicted box, respectively.

Furthermore, EfficientWeld employs a binary cross entropy (BCE) loss function to compute confidence and classification losses:

$$\mathcal{L}_{cls} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (10)$$

where y_i is the ground-truth label and $\hat{y}_i$ denotes the predicted probability of the target class.

One-stage object detection algorithms often suffer from extreme positive-negative sample imbalance, which can degrade network performance. Notably, in this study, the background of weld workpieces is highly complex and cluttered. The target is highly similar to the background, and weld endpoints occupy a very small area—these challenges significantly interfere with target matching. To address these issues, Focal Loss [38] is employed, as shown in Eq. (11):

$$\mathcal{L}_{conf} = \begin{cases} -\alpha (1 - \hat{y}_i) \log(\hat{y}_i), & y = 0 \\ (1 - \alpha) \hat{y}_i \log(1 - \hat{y}_i), & y = 1 \end{cases} \quad (11)$$

where α can increase the weight of positive samples and decrease the weight of the negative samples.

3.2 Weld path generation

3.2.1 Scanning path generation

To generate a welding path, the workpiece must first undergo scanning—requiring prior determination of the scan path. The flowchart of calculating scan path was shown as Fig. 6.

A robot captures an image from a preset position. At this position, the camera's pose in the world coordinate system is represented by matrix M , which corresponds to the camera's extrinsic parameter matrix. For a world coordinate system point $P_w = [x, y, z, 1]$ and its corresponding

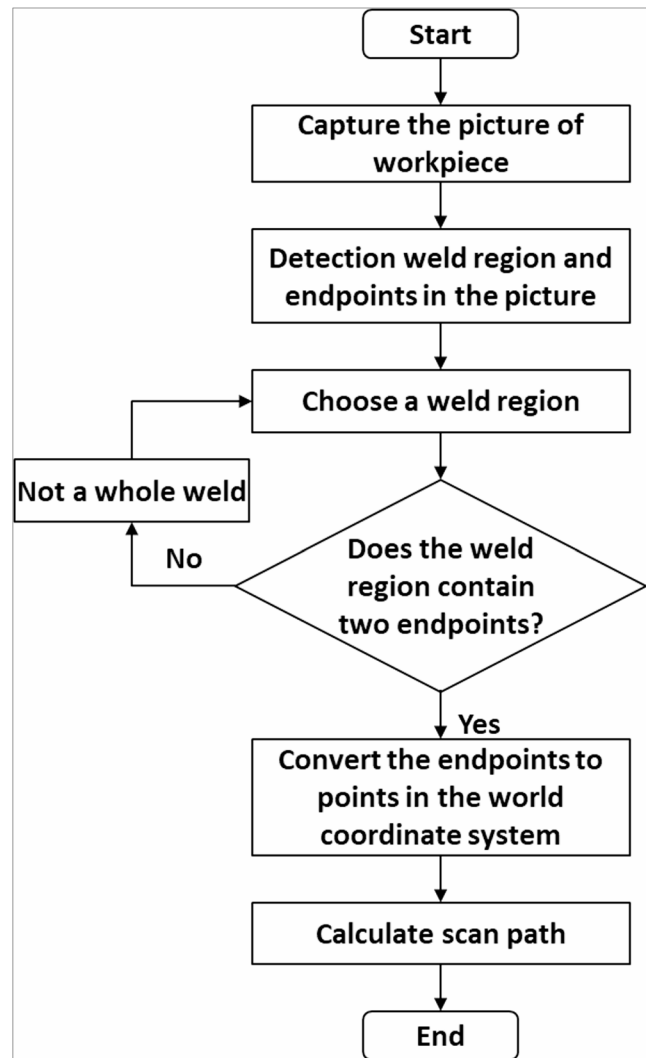


Fig. 6 Flowchart of scan path calculation

camera-frame image coordinates $P_{img} = [u, v, 1]$, the relationship satisfies:

$$z_c P_{img} = A E M P_w \quad (12)$$

where A is the camera's intrinsic parameter matrix.

The image is subsequently processed by the EfficientWeld network (described in Sect. 3.1) to detect weld seams and extract the pixel coordinates of the weld start and end points.

Typically, it is infeasible to directly compute the xyz-coordinates of the weld boundary in the world coordinate system using only the pixel coordinates u and v . However, during world coordinate system setup, the workbench plane is defined as the XOY plane, enabling the Z-coordinate of the weld seam's spatial coordinates to be set to 0. This approximation allows solving for the weld seam's approximate world coordinate position via Eq. (13):

$$\begin{bmatrix} X_W \\ Y_W \\ 1 \\ 1 \end{bmatrix} = {}^W_c T \begin{bmatrix} X_c \\ Y_c \\ 1 \\ 1 \end{bmatrix} \quad (13)$$

where ${}^W_c T = {}^W_E T {}^E_t T {}^t_c T$.

This approximate position is subsequently used to guide a scanner-equipped robot to scan the workpiece, generating a more accurate point cloud model of the weld area.

During scanning, the camera continuously captures images and records its pose (extrinsic parameter matrix) at each capture instant—data later used for point cloud computation.

3.2.2 Center line extraction and point cloud generation

Image processing primarily aims to extract the pixel coordinates of the laser stripe center in the image (Fig. 7). Using light plane calibration results, these pixel coordinates are transformed into world coordinates, thereby generating a point cloud. Since the laser line intensity follows a Gaussian distribution, the grayscale centroid algorithm can extract the laser stripe center. However, this method requires traversing numerous pixel values, significantly reducing computational efficiency.

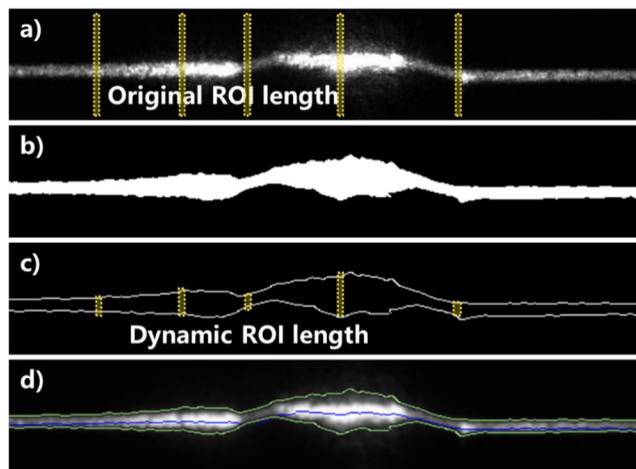
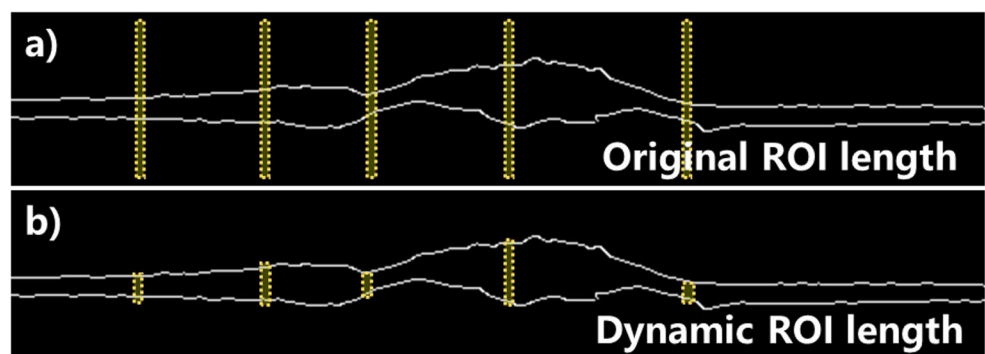


Fig. 7 Original image and processed result. a) Original image; b) Grayscale image; c) Canny edge detection result; d) Extracted center line

Fig. 8 Difference between original ROI and dynamic ROI



To mitigate this limitation, an edge detection step using the Canny operator is incorporated prior to the grayscale centroid algorithm (Fig. 8). The edge-detected region, after appropriate expansion, is defined as the calculation domain for the grayscale centroid algorithm. Additionally, numerous invalid black pixels are efficiently excluded via edge detection, enhancing algorithmic runtime efficiency. The calculation follows:

$$x_u = \frac{\sum_{v=l}^l I(u,v) \cdot v}{\sum_{v=l}^l I(u,v)} \quad (14)$$

where x_u is ordinate value of the laser center point in column u . $I(u,v)$ represents the pixel intensity at (u,v) . l is the length of the ROI which is determined by edge detection.

By fusing the extracted image central points (u,v) with the camera's intrinsic/extrinsic parameters and the light plane equation, the corresponding 3D coordinates of these points are computed. These points form the 3D point cloud of the workpiece.

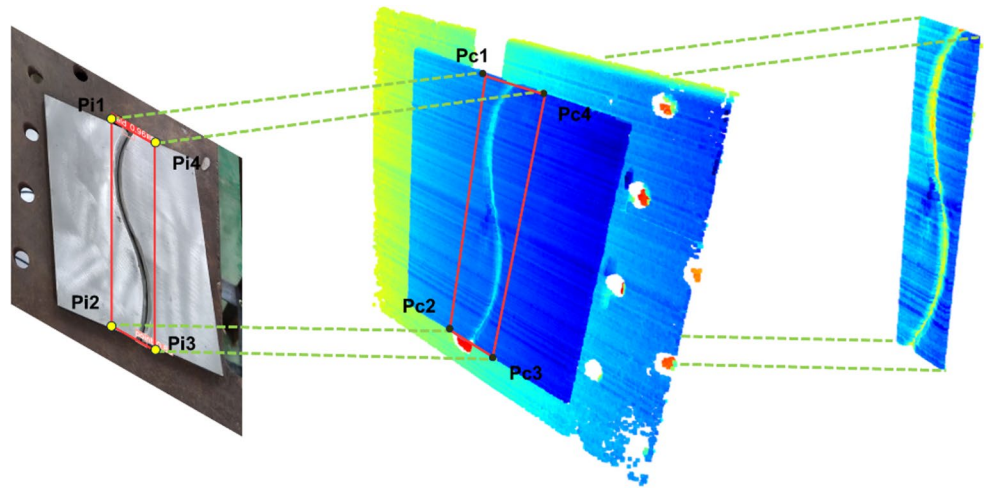
3.2.3 Segmentation of weld seam and path planning

Point clouds acquired in complex industrial environments often contain noise and outliers owing to metal surface light reflection. Consequently, point cloud pre-processing is essential prior to ROI segmentation. During pre-processing, the Statistical Outlier Removal (SOR) method is first used to remove outliers. Subsequently, the Moving Least Squares (MLS) method is applied to enhance point cloud smoothness.

The point cloud extracted from the welding site often has a highly cluttered background, rendering accurate weld seam region extraction challenging.

Figure 9 illustrates the weld seam region extraction method proposed herein, which derives the point cloud weld seam region from the 2D image weld seam region provided by the neural network.

Fig. 9 Schematic diagram of the weld seam region extraction method



$P_{ii}(u_i, v_i)$, $(i = 1, 2, 3, 4)$ denotes the four corner points of the image weld seam region, and $P_{ci}(x_i, y_i, z_i)$, $(i = 1, 2, 3, 4)$ corresponds to these points in the point cloud. Assuming the weld seam region lies in the XOY plane, combining with Eq. (2) yields:

$$\begin{bmatrix} u_i \\ v_i \\ 1 \end{bmatrix} = AEM \begin{bmatrix} x_i \\ y_i \\ 0 \\ 1 \end{bmatrix} \quad (15)$$

By solving for the four P_{ci} points, the region bounded by connecting these points is identified as the weld seam region.

After acquiring the workpiece point cloud, further segmentation is required to isolate the weld area. The weld area is segmented by identifying the minimum Z-value. Given the concave nature of the weld area, all point Z-coordinates are sorted, and the first K points are selected to define the weld area. Finally, a welding path comprising K points is generated.

The welding torch angle is a critical parameter in welding processes, significantly influencing welding quality. Thus, the welding path alone is insufficient; the welding torch posture must also be considered during welding. The posture can be represented by the approach vector (v_{ap}) and attitude vector (v_{at}), corresponding to the Z-axis and X-axis of the tool coordinate system O_t , respectively.

During welding, the welding torch should be perpendicular to the workpiece in the Z-direction. Plane fitting is performed on the segmented point cloud to obtain the plane equations:

$$Ax + By + Cz + D = 0 \quad (16)$$

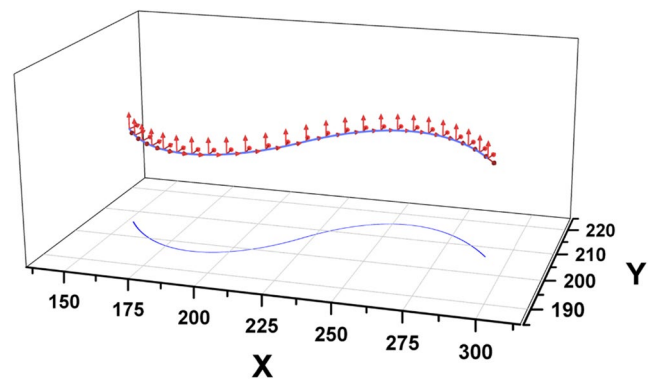


Fig. 10 Planned welding path and corresponding welding torch postures

Thus,

$$v_{ap} = \frac{(A, B, C)}{\|(A, B, C)\|} \quad (17)$$

In the XY-direction, considering wire feeding requirements during welding, the feeding mechanism should be parallel to the weld seam tangent direction:

$$v_{ati} = \begin{cases} \frac{\overline{P_1 P_2}}{\|\overline{P_1 P_2}\|}, & i = 1 \\ \frac{\overline{P_{i-1} P_{i+1}}}{\|\overline{P_{i-1} P_{i+1}}\|}, & i = 2, 3, \dots, K-1 \\ \frac{\overline{P_{K-1} P_K}}{\|\overline{P_{K-1} P_K}\|}, & i = K \end{cases} \quad (18)$$

where i is the path point index, and P_i denotes the coordinates of the i -th path point.

The Y-axis direction of the tool coordinate system O_t is calculated as $v_{ati} \times v_{ap}$. Figure 10 shows the planned welding path and the corresponding welding torch postures.

4 Experiments and results

4.1 Weld seam detection

4.1.1 Data collection and augmentation

Considering the requirement for diverse weld seams in practical engineering applications, 222 original images of S-shape butt weld, linear butt weld and lap weld were taken using the camera in Sect. 2.1.

To enhance network robustness, data augmentation algorithms, including image flipping, noise injection, brightness/contrast adjustment, and image blurring, were employed to expand the dataset size. Figure 11 illustrates the performance of these algorithms, and 1,998 images were generated by randomly combining these algorithms applied to the original images. Finally, a dataset of 2,220

images was split into training and validation subsets at an 8:2 ratio.

4.1.2 Network training

Due to the small size of the weld dataset, the COCO dataset was used for pre-training to achieve better model performance. Using pre-trained weights as initial weights (instead of random initialization) accelerates convergence, reduces training time, enhances generalization ability, and mitigates overfitting, particularly for limited datasets, thereby improving data utilization efficiency. The proposed EfficientWeld network was developed upon the YOLOv5 framework. Thus, the YOLOv5 model (n, s, and m) was selected for comparison with our method.

The training and validation results of EfficientWeld models on the self-constructed dataset are presented in Fig. 12a) and b); The Precision-epoch, Recall-epoch and

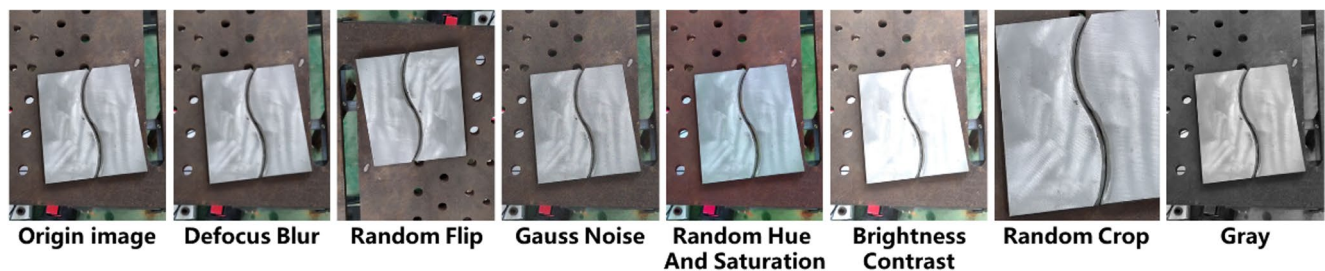
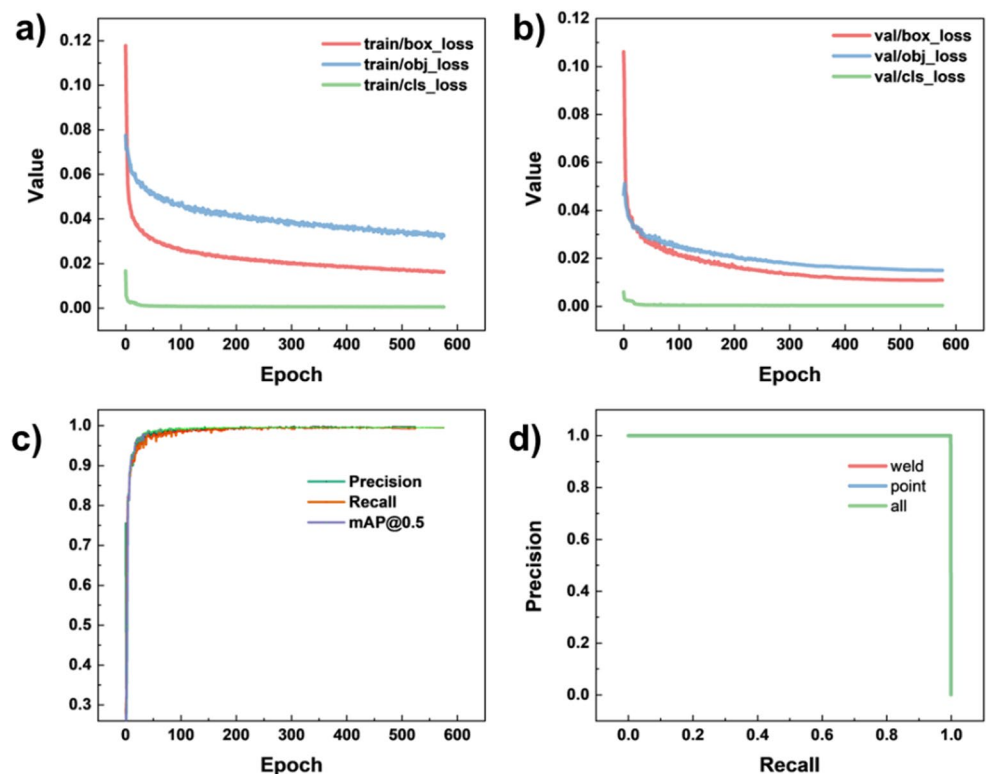


Fig. 11 Different data augmentation algorithms

Fig. 12 Training results. a) Loss curves of training subsets. b) Loss curves of validation subsets. c) Precision-epoch, Recall-epoch and mAP@0.5-epoch curves of training subsets. d) PR curves of training subsets



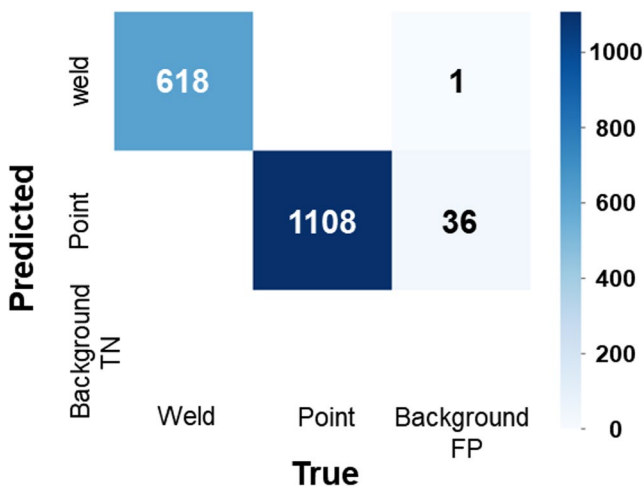


Fig. 13 Confusion matrices of validation subsets

mAP@0.5-epoch curves of training subsets are presented in Fig. 12c); The PR curves are presented in Fig. 12d). The confusion matrix result is presented in Fig. 13. In this study, precision, recall, and mAP (mean Average Precision) at different IoU thresholds were used to assess the reliability of two models. Precision is defined as the ratio of correctly predicted positive instances to the total number of predicted positive instances. Recall is defined as the ratio of correctly predicted positive instances to the total number of actual positive instances. IoU quantifies the overlap degree between two bounding boxes (predicted vs. ground-truth). It is calculated as the ratio of the intersection area to the union area of the two boxes, ranging from 0 to 1—higher values indicate a better match. Typically, an IoU threshold (e.g., 0.5) is set to classify predictions as “true positives” (TPs). Most object detection models, including ours, use an IoU threshold of 0.5 as the performance evaluation baseline.

While precision and recall are key metrics, they only reflect model behavior under specific conditions. mAP (mean Average Precision), by contrast, offers a more comprehensive evaluation across varying thresholds, classes, and IoU criteria—it is a critical metric for assessing object detection accuracy. Two common mAP standards are mAP@0.5 (IoU threshold=0.5) and mAP@0.5:0.95 (average mAP across IoU thresholds 0.5–0.95, step 0.05), the latter providing stricter bounding box accuracy evaluation.

Table 2 presents the performance test results of different models. EfficientWeld exhibited nearly identical precision and mAP@0.5 scores compared to YOLOv5m, while achieving the highest recall score. Meanwhile, YOLOv5m attained the highest mAP@0.5:0.95 score. Beyond precision, efficient inference speed is critical for object detection. The inference time for weld seam detection was measured for both YOLOv5 and EfficientWeld using an NVIDIA GTX 1650Ti GPU. As shown in Table 5, YOLOv5m requires 28.9 ms to detect weld seams in a single image. EfficientWeld, by comparison, requires only 7.8 ms—an improvement of 73% over the YOLOv5m and 36% of YOLOv5s. Given the application scenario of this study, the model’s primary task is to provide approximate start and end points locations for guiding subsequent scanning operations, as precise weld seam positions are instead derived from subsequent point clouds. Accordingly, the successful identification of weld seam endpoints holds greater criticality than localization accuracy. Therefore, although EfficientWeld demonstrates a lower mAP score, it remains the optimal choice among the aforementioned models.

Model output results are presented in Fig. 14. In Fig. 14(a), the locations of two complete welds and their endpoints are successfully identified in a single image. In Fig. 14(b), a linear weld’s position is identified; however, since it is incomplete in the image, its endpoints are not detected. For the S-shaped weld, both the weld position and endpoints are identified. Figure 14(c) shows a blurred image captured in a defocused state—despite the blur, the model still identifies the processed weld’s location and its two endpoints, demonstrating robustness.

4.2 Efficiency analysis of proposed method

The processing efficiency of the weld detection and path planning algorithm is also a critical factor for evaluating its advantages over alternatives. For the two weld workpieces used in this experiment, the time required to generate weld trajectories was recorded, with a total of 42.23 s. Table 3 presents the time consumed by each stage. The most time-consuming step is 3D reconstruction (implemented in Python) and rewriting the program in C/C++ could achieve faster execution speeds. In comparison, manual teaching

Table 2 Performance of EfficientWeld and YOLOv5 model

	Precision	Recall	mAP @0.5	mAP @0.5:0.95	FLOPs	Inference Time
EfficientWeldn	0.997	0.994	0.995	0.857	1.9 G	4.3 ms
EfficientWeld	0.999	0.996	0.995	0.906	5.6 G	7.8 ms
YOLOv5n	0.995	0.995	0.995	0.896	4.1 G	5.2 ms
YOLOv5s	0.997	0.995	0.995	0.931	15.8 G	12.2 ms
YOLOv5m	0.999	0.995	0.995	0.950	47.9 G	28.9 ms

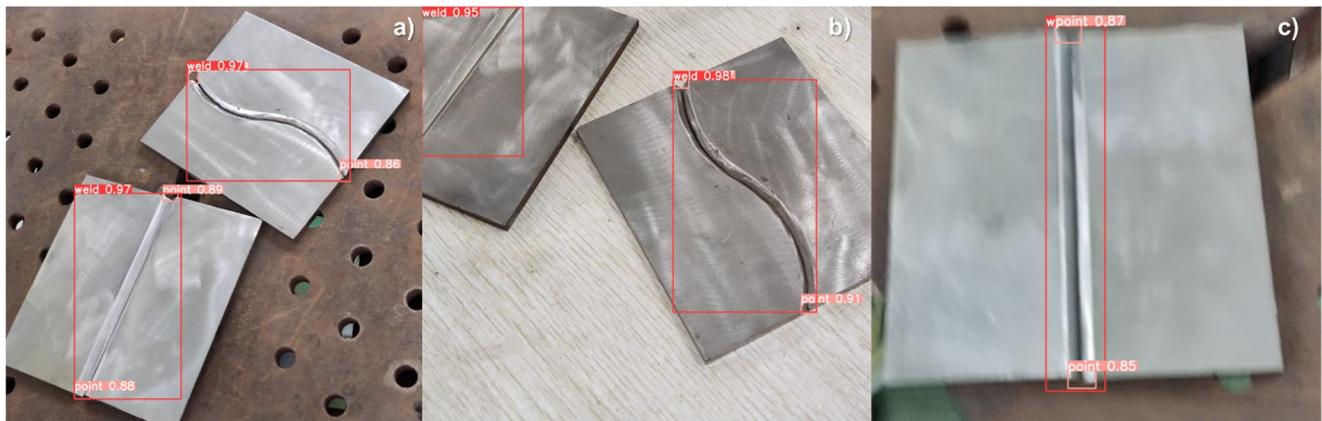


Fig. 14 Identification results for different weld images. a) Different weld types; b) Incomplete weld; c) Unclear weld

Table 3 Time consumed by each step

Steps	Running time
Scan and reconstruction	41.03s
Point cloud segmentation	0.75s
Feature extraction	0.42s
Path fitting	0.03s
Total	42.23s

of this weld seam takes approximately 5–10 min. The proposed method meets industrial production requirements.

4.3 Error analysis of proposed method

As shown in Fig. 15, the accuracy of extracted weld paths was verified by comparing with the “teach-playback” method. Table 4 presents the calculated extracted length error, transverse maximum error (ME), and transverse root mean square error (RMSE). The proposed method yields

ME values of less than 0.61 mm and RMSE values of less than 0.25 mm for both weld paths, meeting welding process error requirements.

4.4 Weld experiment

The actual welding was carried out using the method proposed in this study, and the welding process parameters are shown in the Table 5.

A 5 mm-thick 316 stainless steel plate was employed, with a groove angle of 45°, a root face thickness of 2 mm, and 316 L stainless steel welding wire as the filler material.

The actual welding effect is presented in Fig. 16. For the TIG welding process with a relatively low deposition rate, the forming effects of the first weld pass were compared. Figure 16 (a) and (b) present the first weld passes fabricated using the teach-playback method and the method proposed in this study, respectively. Both exhibited relatively good

Fig. 15 Weld position error extracted by our method. a) Line weld; b) S-Line weld

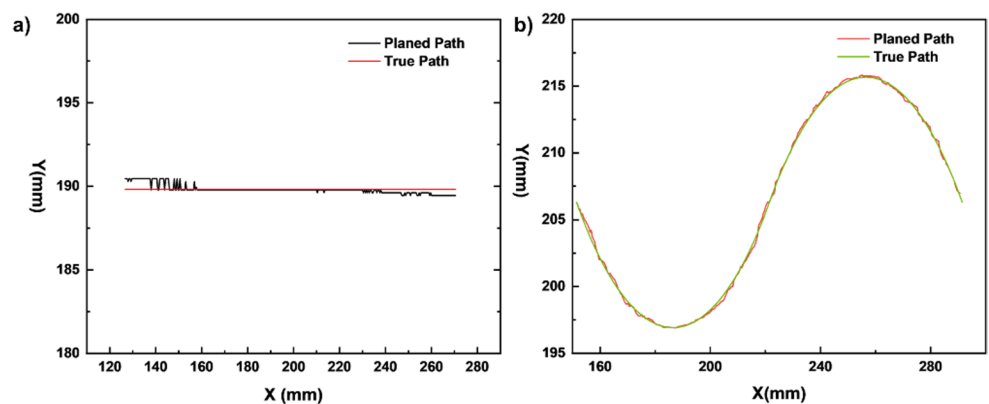
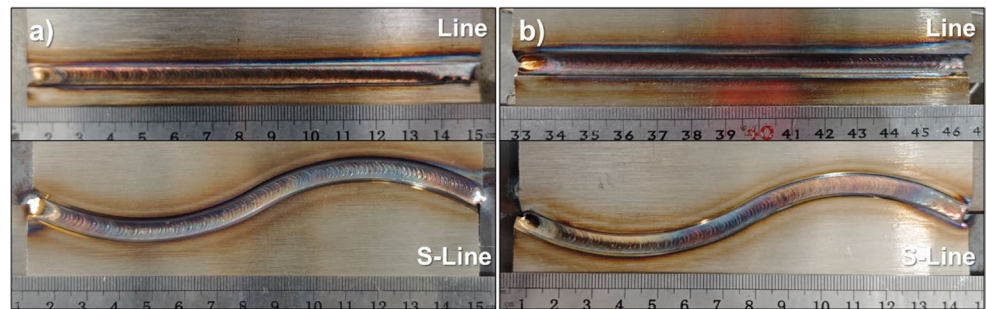


Table 4 Weld position error metrics

	Scan length (mm)	True length (mm)	Length error (mm)	Max error (transverse) (mm)	RMSE (transverse) (mm)
Line	142.99	141.21	1.78	0.58	0.25
S-Line	139.98	138.47	1.51	0.61	0.19

Table 5 Welding experiment parameters

Current	Frequency	Speed	Electrode diameter	Wire diameter	Wire feed speed	Shielding gas
Peak: 150 A Base: 90 A	5 Hz	2 mm/s	2.4 mm	1.6 mm	0.8 m/min	Ar, 12 L/min

Fig. 16 Appearance morphology of the weld. a) “teach-playback” mode; b) method proposed in this study

forming quality. The primary distinction between the two approaches lies in the time required for path planning. For the proposed method, weld shape had no impact on time consumption, both welds required 40–50 s. In contrast, for the teach-playback method, greater weld geometry complexity required more teaching points, resulting in longer preparation time. In this experiment, straight weld seams only required programming of the start and end points (approximately 1 min). However, curved weld seams demanded at least 5 teaching points and adjustment of the wire feed direction at each point, requiring approximately 8 min. Therefore, the method proposed herein can significantly improve operational efficiency.

5 Conclusion

From an industrial application perspective, this study proposes an innovative automated robotic welding path planning framework based on deep learning and 3D point cloud segmentation. Additionally, a robot programming system was developed, integrating weld seam recognition, point cloud acquisition, segmentation, and feature extraction. The main conclusions of this study are as follows:

- 1) The proposed method adopts a coarse-to-fine strategy, ensuring both the accuracy and efficiency of weld seam extraction. By leveraging artificial intelligence to identify weld seam endpoint positions, the scanner is guided to perform localized fine scanning, thereby overcoming the limitations of global scanning, such as large data volumes and difficulties in feature extraction.
- 2) The EfficientWeld network can quickly and accurately extract weld seams and their endpoints. The model adapts to variations in the welding environment and accurately identifies lap welds, butt welds, and irregular welds located in diverse backgrounds.

- 3) Based on point cloud data, the welding path and torch posture were systematically planned. The generated path achieves a root-mean-square error (RMSE) of <0.25 mm and a maximum lateral error of <0.61 mm, meeting practical application requirements. For S-Line curved weld seams, the path generation speed is significantly faster than that of the “teach-playback” programming method.

Regarding future works, several limitations of this study can be addressed. First, EfficientWeld is not applicable to scenarios where the weld is too long to be fully captured or exhibits a closed annular configuration. For welds exceeding the camera’s field of view, adjustments including repositioning the robotic camera or substituting it with a short-focal-length lens can ensure the entire workpiece is captured. Closed annular structures pose a challenge to this method due to the lack of the explicit start/end features required by EfficientWeld for recognition. Future work may address this limitation through additional datasets and incremental learning. Second, this scheme is not applicable to automatic path planning for multi-layer and multi-pass welding. It was designed for single-layer welds with a thickness under 6 mm. For thicker plates, larger groove dimensions are required during welding. Consequently, further geometric analysis of the segmented point cloud is necessary to enable path planning for multi-layer and multi-pass welding.

Author contributions Chengyuan Ma: coding, data collecting, data analysis, investigation, visualization, and writing—original draft. Zheng Meng: coding, data collecting, visualization. Zhaochuan Wang: conceptualization, investigation, visualization. Shenghong Yan: conceptualization, editing. Bo Chen: methodology, resources, supervision, and writing—original draft. Caiwang Tan: conceptualization, supervision. Xiaoguo Song: supervision.

Funding This research was supported by the National Natural Science Foundation of China (No. 52175309), the Taishan Scholars Program (No. tsqn202312323), State Key Laboratory of Precision Welding & Joining of Materials and Structures (No. 24-T-07).

Declarations

Competing interests The authors have no competing interests to declare that are relevant to the content of this article.

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